

UNIT 6 : STABILITY OF SYSTEMS

1. Proportional Controller :

⇒ Proportional Controller produces an output which is proportional to Error signal

$$\therefore u(t) \propto e(t)$$

$$\therefore u(t) = k_p \cdot e(t) \quad - (1)$$

Where, k_p = proportional constant.

Taking Laplace transform on both sides, we get ;

$$\therefore U(s) = k_p \cdot E(s)$$

$$\therefore \frac{U(s)}{E(s)} = k_p$$

$$\therefore T.F = \boxed{\frac{U(s)}{E(s)} = k_p} \quad - (2)$$

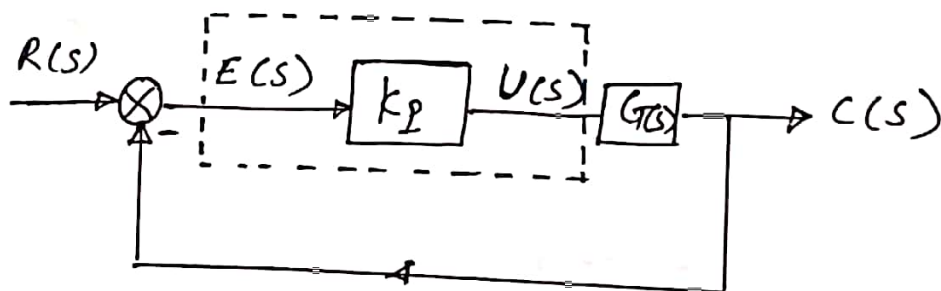


Fig: Block-diagram proportional Controller.

$R(s) \rightarrow$ reference i/p.

$C(s) \rightarrow$ Controlled o/p.

$E(s) \rightarrow$ Laplace of error signal $e(t)$.
 $U(s) \rightarrow$ Laplace of o/p $u(t)$
 $G(s) \rightarrow$ Equivalent forward path Transfer function.

2. Derivative Controller :

$\Rightarrow$ Derivative Controller produces an output which is derivative of Error signal.

$$\therefore u(t) \propto \frac{de(t)}{dt}$$

$$\therefore u(t) = k_D \cdot \frac{d}{dt} e(t) \quad - (1)$$

Where, k_D = Derivative Constant.

Taking Laplace transform on both sides, we get;

$$U(s) = k_D \cdot s E(s)$$

$$\therefore \boxed{\frac{U(s)}{E(s)} = k_D \cdot s} \quad - (2)$$

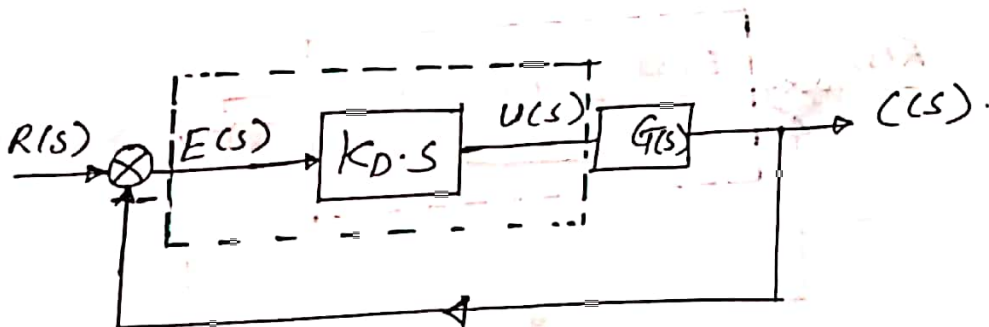


Fig: Block diagram derivative Controller.

3.

Integral Controller :

⇒ Integral Controller produces an output which is integral of the error signal.

$$\therefore u(t) \propto \int e(t)$$

$$\therefore u(t) = K_I \int e(t) \quad \text{--- (1)}$$

where, K_I = Integral Constant

Taking L.T on Both sides;

$$U(s) = K_I \cdot \frac{E(s)}{s}$$

$$\therefore T.F = \frac{U(s)}{E(s)} = \frac{K_I}{s}$$

$$T.F. = \boxed{\frac{U(s)}{E(s)} = \frac{K_I}{s}} \quad \text{--- (2)}$$

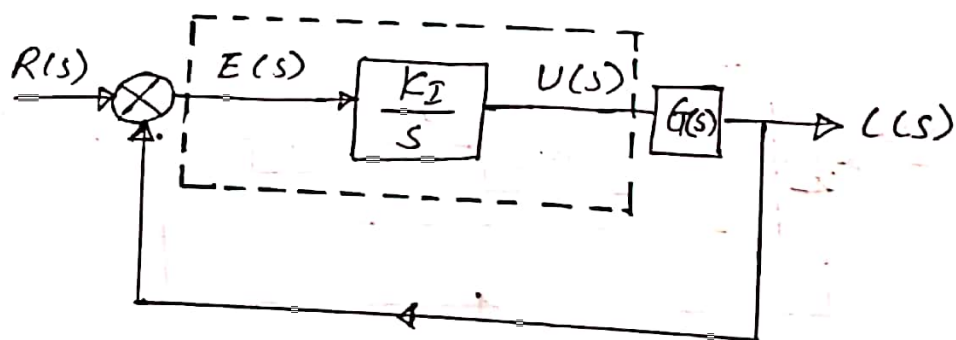


fig: Block diagram integral Controller.

4. PI Controller : (proportional integral Controller)

⇒ It produces an output which is the combination of output of proportional and Integral Controller.

$$\therefore u(t) = K e(t) + \int e(t)$$

$$\therefore u(t) = K_p e(t) + K_I \int e(t) \quad - (1)$$

where, K_p = proportional constant
& K_I = Integral constant.

Taking L.T on both sides.

$$U(s) = K_p E(s) + K_I \cdot \frac{E(s)}{s}$$

$$= E(s) \left[K_p + K_I \cdot \frac{1}{s} \right]$$

$$\therefore U(s) = E(s) \left[K_p + \frac{K_I}{s} \right]$$

$$\therefore \frac{U(s)}{E(s)} = K_p + \frac{K_I}{s}$$

$$\therefore T.F = \boxed{\frac{U(s)}{E(s)} = K_p + \frac{K_I}{s}} \quad - (2)$$

$$= K_p \left(1 + \frac{1}{T_I \cdot s} \right) \left[\because T_I = \frac{K_p}{K_I} \right]$$

$$\therefore \boxed{\frac{U(s)}{E(s)} = K_p \cdot \left(1 + \frac{1}{T_I \cdot s} \right)} \quad - (3)$$

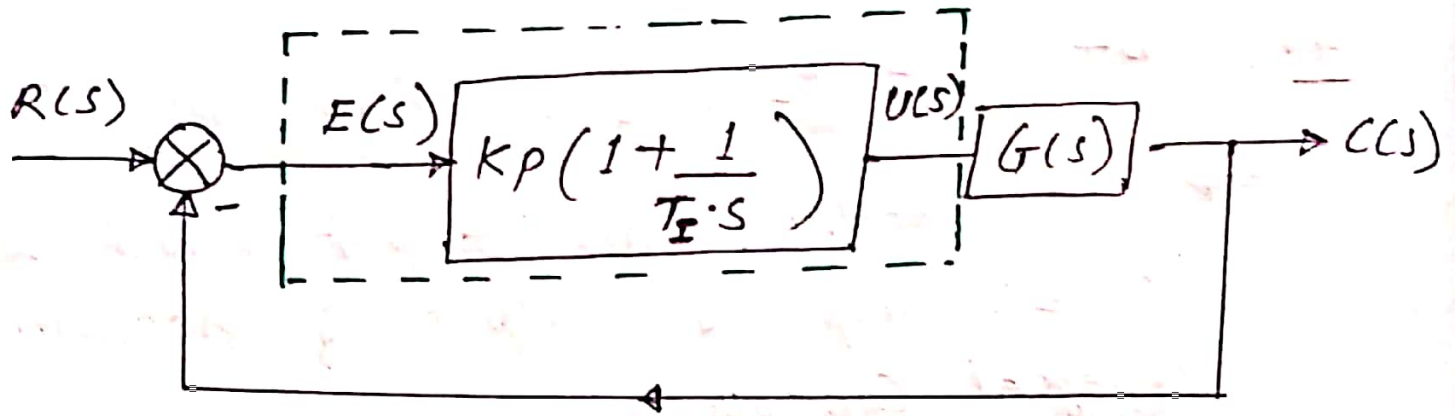


fig: Block-diagram PI Controller

5. PD Controller : (Proportional Derivative Controller)

⇒ It produces an output, which is the combination of the outputs of proportional and derivative controller.

$$\therefore u(t) \propto e(t) + \frac{d}{dt} \cdot e(t)$$

$$\therefore u(t) = K_p \cdot e(t) + K_D \cdot \frac{d}{dt} e(t) \quad - (1)$$

Where;

K_p = proportional constant

K_D = derivative constant.

Taking L.T on both sides, we get;

$$\therefore U(s) = K_p \cdot E(s) + K_D \cdot s E(s)$$

$$\therefore U(s) = E(s) (K_p + K_D \cdot s)$$

$$\therefore \boxed{\frac{U(s)}{E(s)} = K_p + K_D \cdot s} \quad - (2)$$

Eqn (2) can be written as,

$$\frac{U(s)}{E(s)} = K_p \left(1 + \frac{K_D}{K_p} \cdot s \right)$$

$$\therefore TF = \boxed{\frac{U(s)}{E(s)} = K_p (1 + T_D \cdot s)} \quad - (3) \quad \left(\because T_D = \frac{K_D}{K_p} \right)$$

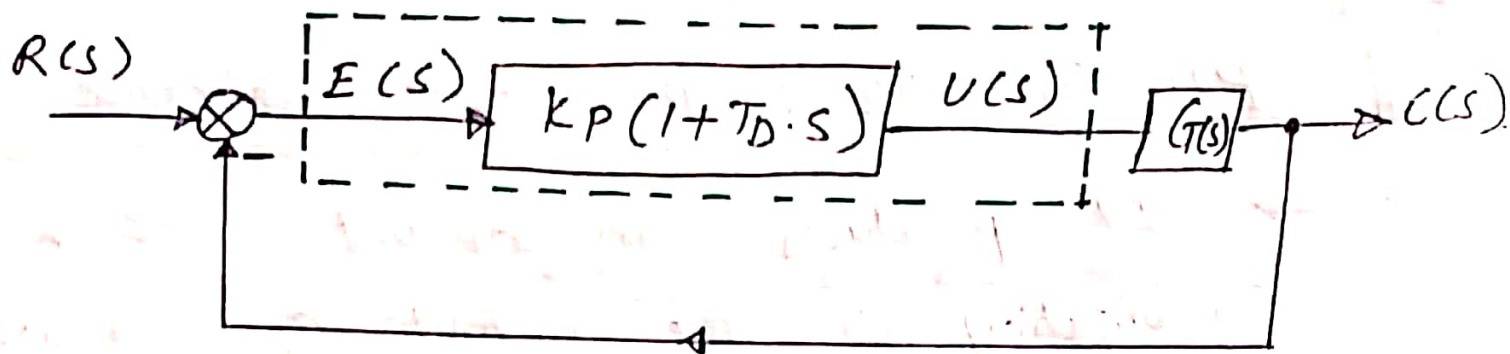


fig: Block-diagram PD Controller.

[6]. PID Controller : (Proportional - Integral - Derivative Controller)

⇒ It produces an output, which is the combination of the outputs of proportional, integral & derivative controllers.

$$\therefore u(t) \propto e(t) + \int e(t) + \frac{d}{dt} e(t)$$

$$\therefore u(t) = K_P \cdot e(t) + K_I \cdot \int e(t) + K_D \cdot \frac{d}{dt} e(t) \quad - (1)$$

Where; K_P = proportional constant

K_I = Integral constant

K_D = derivative constant.

Taking L.T on both sides,

$$\begin{aligned} U(s) &= K_P \cdot E(s) + \frac{K_I}{s} \cdot E(s) + K_D \cdot sE(s) \\ &= E(s) \cdot \left(K_P + \frac{K_I}{s} + K_D \cdot s \right) \end{aligned}$$

$$\therefore \boxed{\frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D \cdot s} \quad - (2)$$

∴ on further simplification eqⁿ (2) becomes

$$= K_P \left[1 + \frac{K_I}{K_P \cdot s} + \frac{K_D}{K_P} \cdot s \right]$$

$$= K_P \left[1 + \frac{1}{T_I \cdot s} + T_D \cdot s \right]$$

$$\therefore \frac{U(s)}{E(s)} = K_p \left[1 + \frac{1}{T_I \cdot s} + T_D \cdot s \right] \quad - (3)$$

Where,

$$T_I = \frac{K_p}{K_I} \quad \& \quad T_D = \frac{K_D}{K_p}$$

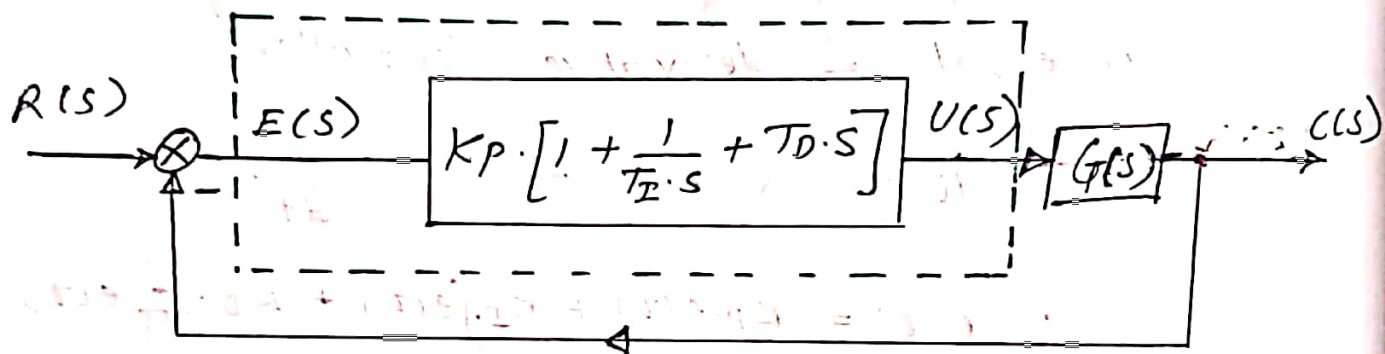


Fig: Block diagram PID Controller

① Controller : (Defn)

⇒ A Controller is basically "a unit present in a control system that generates control signals to reduce the deviation of the actual value from the desired value to almost zero or lowest possible value."

Ex :- A.C $\left\{ \begin{array}{l} \text{Actual value} \rightarrow \text{atm. temp} \\ \text{Desired value} \rightarrow \text{set. temp} \end{array} \right\}$

<1> Proportional Controller :

⇒ The proportional Controller produces an o/p, which is proportional to error signal $[e(t)]$.

<2> Integral Controller : (Corrects errors) .

→ Integral Controller reduces the error to zero, thereby eliminating the problem with proportional Controller.

<3> Derivative Controller : (Maintains stability)

→ Derivative Controller, detects and resists abrupt (sudden) changes in the system.